MIL-C-47121 (MI)
24 May 1974
SUPERSEDING
MIS-13923A
30 January 1967

MILITARY SPECIFICATION

COMPOUND, SEALING, POLYSULFIDE RUBBER

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a polysulfide rubber compound for sealing electrical and mechanical components as a protection against corrosion, moisture, and mechanical damage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/5	Aluminum Alloy, Alclad 2024, Plate and Sheet
QQ-B-613	Brass, Leaded and Non-leaded: Flat Products (Plate, Bar, Sheet, and Strip)
TT-T-548	Toluene, Technical

Military

MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

FSC 8030

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Tests for.
ASTM D 150	A-C Loss Characteristics and Dielectric Constant Permittivity of Solid Electrical Insulating Materials, Tests for.
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Tests for.
ASTM D 903	Peel or Stripping Strength of Adhesives Bonds, Test for.
ASTM D 1706	Method of Test for Indentation Hardness of Plastics by Means of a Durometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

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Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington, D.C. 20013.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Materials.

3.2.1 Composition and condition in container.

3.2.1.1 Composition. The sealing compound shall consist of 2 components: a base material and a separate curing agent. The basic ingredient of the material shall be an organic polysulfide liquid polymer which shall constitute at least 55 percent by weight of the cured material. The base material shall be cured by the addition of the curing agent and shall not depend upon solvent evaporation for setting. Benzene or other highly toxic materials shall not be present either initially or as a product of the curing reaction.

3.2.1.2 Condition in container. The components of the compound shall be homogeneous, and free of gel particles, dirt, or any other contaminant which would affect the intended use of the compound.

3.2.2 Viscosity of base material. The base material shall have a viscosity between 8,000 and 14,000 poises.

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3.2.3 Extrusion rate. The material shall have an extrusion rate of not less than 12.0 grams per minute.

3.2.4 Storage life. The storage life of the base material and the curing agent shall be a minimum of 6 months stored between 60 and 90 degrees Fahrenheit (F) (16 and 32 degrees Celsius (C)).

3.3 Performance Characteristics.

3.3.1 Hardness. The Shore A hardness of the cured compound shall be from 30 to 60.

3.3.2 Peel strength. The initial breakaway peel strength shall be not less than 18 pounds per inch width and the average peel strength shall be not less than 14 pounds per inch width.

3.3.3 Tensile shear strength. The average tensile shear strength of bonded specimens shall be not less than 200 pounds per square inch (psi), with the lowest value for an individual specimen not less than 150 psi.

3.3.4 Dielectric strength. The material shall have a dielectric strength of not less than 250 volts per mil.

3.3.5 Dielectric constant. The material shall have a maximum dielectric constant of 11.5.

3.3.6 Volume resistivity. The material shall have a volume resistivity of not less than 10^{11} ohm-centimeters.

3.4 Environmental condition.

3.4.1 Resistance to thermal degradation. The material shall be capable of withstanding exposure up to 250 degrees F (121 degrees C) for 48 plus or minus 1 hour, without swelling, blistering, or internal foaming. After exposure, the average tensile shear strength shall be not less than 200 psi with no individual specimen below 150 psi.

3.5 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in conformance with this specification. The cured compound shall show no evidence of discoloration, internal foaming, or external cracking.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. When specified in the contractual document (see 6.2), a sample consisting of 1 pound of resin and 2 ounces of curing agent, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table I, shall be conducted in accordance with the corresponding test paragraphs.

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Table I. Preproduction inspection

Characteristic	Requirement source	Test paragraph
Composition	3.2.1.1	4.5.4.1
Condition in container	3.2.1.2	4.5.4.1
Viscosity of base material	3.2.2	4.5.4.2
Extrusion rate	3.2.3	4.5.4.3
Storage life	3.2.4	4.5.4.4
Hardness	3.3.1	4.5.4.5
Peel strength	3.3.2	4.5.4.6
Tensile shear strength	3.3.3	4.5.4.7
Dielectric strength	3.3.4	4.5.4.8
Dielectric constant	3.3.5	4.5.4.9
Volume resistivity	3.3.6	4.5.4.10
Resistance to thermal degradation	3.4.1	4.5.4.11
Workmanship	3.5	4.5.5

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4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table II, shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table II. Acceptance inspection

Characteristic	Requirement source	Test and inspection paragraph
Condition in container	3.2.1.2	4.5.4.1
Extrusion rate	3.2.3	4.5.4.3
Hardness	3.3.1	4.5.4.5
Peel strength	3.3.2	4.5.4.6
Workmanship	3.5	4.5.5

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a. Temperature, room ambient plus or minus 9 degrees C (plus or minus 16 degrees F)

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b. Altitude, facility ground

c. Humidity, facility ambient up to 95 percent relative humidity

4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Test specimen preparation.

4.5.3.1 Mixing. The sealing base material and its curing agent, both in the original unopened containers, together with the mixing equipment, shall be held at a temperature of 75 degrees plus or minus 2 degrees F (24 degrees plus or minus 1 degree C) for not less than 4 hours. The base material and the curing agents shall be mixed thoroughly in the proportions recommended by the manufacturer. The mixing operation shall be conducted at a temperature of 75 degrees plus or minus 2 degrees F (24 degrees plus or minus 1 degree C) for not less than 3 minutes.

4.5.3.2 Application of sealing compound. All test specimens shall be thoroughly cleaned with toluene conforming to TT-T-548 prior to the application of the sealing compound. The sealing compound shall be applied by an extrusion gun, or equivalent, taking care to avoid the entrapment of air by a folding action.

4.5.3.3 Curing. Test specimens shall be cured in a circulating air oven at 125 degrees plus or minus 5 degrees F (52 degrees plus or minus 3 degrees C) for 4 hours plus or minus 10 minutes. Upon the completion of the curing period the specimens shall be removed from the oven and allowed to come to room temperature. All tests shall be conducted not less than 2 hours following the completion of the above cure.

4.5.4 Tests.

4.5.4.1 Composition and condition in container. Composition shall be certified by the supplier and the condition in container shall be examined visually to establish compliance to 3.2.1.

4.5.4.2 Viscosity. The viscosity of the base material shall be measured with a Brookfield viscosimeter, Model RVF, or equivalent, as follows:

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Spindle	Speed
7	2 revolutions per minute (rpm)

The base material and the viscosimeter shall be at uniform temperature of 75 degrees plus or minus 2 degrees F (24 degrees plus or minus 1 degree C) during the test. The base material shall be thoroughly stirred with a spatula immediately before testing. Readings shall be taken when the pointer first assumes a steady position after the release of the clutch. The viscosity shall conform to 3.2.2.

4.5.4.3 Extrusion rate. Freshly mixed material shall be poured into a polyethylene extrusion cartridge fitted with a nozzle having an orifice diameter of 1/8 inch. The cartridge shall be installed in an extrusion gun and connected to a 60 plus or minus 2 psi source of compressed air. The rate of extrusion shall be measured 30 minutes after mixing and expressed as grams per minute, and shall conform to 3.2.3.

4.5.4.4 Storage life. After storage under conditions specified in 3.2.4 the materials shall be tested to determine compliance with the characteristics listed in Table I in accordance with the corresponding test paragraphs. The supplier's certification of compliance certifying to the storage life requirements (see 3.2.4) may be accepted during the storage period.

4.5.4.5 Hardness. The hardness of a specimen having the minimum dimensions of 2 inches by 2 inches by 1/4 inch in thickness shall be determined in accordance with ASTM D 1706. Readings shall be made 10 seconds after firm contact has been established. The hardness shall conform to 3.3.1.

4.5.4.6 Peel strength. The peel strength of the sealant shall be determined in accordance with ASTM D 903 with the exception that a brass strip conforming to QQ-B-613, Composition 1, annealed, measuring 0.50 inch by 8.00 inches by 0.004 to 0.005 inch in thickness shall be used as the flexible member. The rigid member shall be an aluminum strip of the same size conforming to QQ-A-250/5 and 0.064 inch thick. The rigid member shall be anodized in accordance with MIL-A-8625, Type I. The peel rate shall be 12 inches per minute. The peel strength shall conform to 3.3.2.

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4.5.4.7 Tensile shear strength. Aluminum test strips conforming to QQ-A-250/5 and measuring 1.00 plus or minus 0.005 inch by 8.00 plus or minus 0.010 inch by 0.064 inch stock thickness shall be prepared and anodized in accordance with MIL-A-8625, Type I. Five pairs of these specimens shall be bonded together with the sealing material by overlapping the ends 0.50 plus or minus 0.010 inch. The tensile shear strengths of the bonded joints shall be determined in a suitable testing machine at room temperature and a strain rate of 0.25 inch per minute, and shall conform to 3.3.3.

4.5.4.8 Dielectric strength. Sheets of sealing material approximately 0.015 inch in thickness shall be prepared and tested as specified in ASTM D 149. The dielectric strength shall conform to 3.3.4.

4.5.4.9 Dielectric constant. Discs of sealing material approximately 2 inches in diameter and 1/8 inch in thickness shall be prepared and tested as specified in ASTM D 150. The dielectric constant shall conform to 3.3.5.

4.5.4.10 Volume resistivity. Discs of the sealing material approximately 2 inches in diameter and 1/8 inch in thickness shall be prepared and tested as specified in ASTM D 257. The volume resistivity shall conform to 3.3.6.

4.5.4.11 Resistance to thermal degradation. Tensile shear strength specimens shall be prepared as specified in 4.5.4.7. The specimens shall be placed in an air circulating oven for 48 hours plus or minus 1 hour at 250 degrees plus or minus 10 degrees F (121 degrees plus or minus 5.5 degrees C). The specimens shall be tested for tensile shear strength as specified in 4.5.4.7. The sealing material shall be examined for swelling, blistering, or internal foaming, as specified in 4.5.5. The resistance to thermal degradation shall conform to 3.4.1.

4.5.5 Workmanship. The material shall be visually examined for conformance to workmanship requirements specified in 3.5. The specimen used for the hardness measurement or a similar specimen shall be visually examined for cracks and discoloration. The specimen shall then be bisected with a sharp knife and the interior shall be examined for discoloration, swelling, blistering, and foaming. The workmanship shall conform to 3.5.

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4.6 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Unless otherwise specified, packaging shall be Level C.

5.1.1 Level C. Material shall be packaged to afford adequate protection against deterioration and damage during shipment from supply source to the first receiving activity. The curing agent shall be packaged with base compound to prevent accidental separation, but in a manner to provide easy separation for mixing purposes.

5.2 Packing. Unless otherwise specified, packing shall be Level C.

5.2.1 Level C. Unit packages shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Rules as applicable.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The materials in accordance with this specification are intended for potting and sealing electronic components and for sealing mechanical parts to protect against corrosion or contamination by excluding water, aircraft fluids, gases, dust, and foreign objects. They may also be used to effect a hermetic seal for electrical and mechanical components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Requirement for certification of composition (see 4.5.4.1).

d. Quantity and size of container.

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6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13923A, dated 30 January 1967.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 8030-A049

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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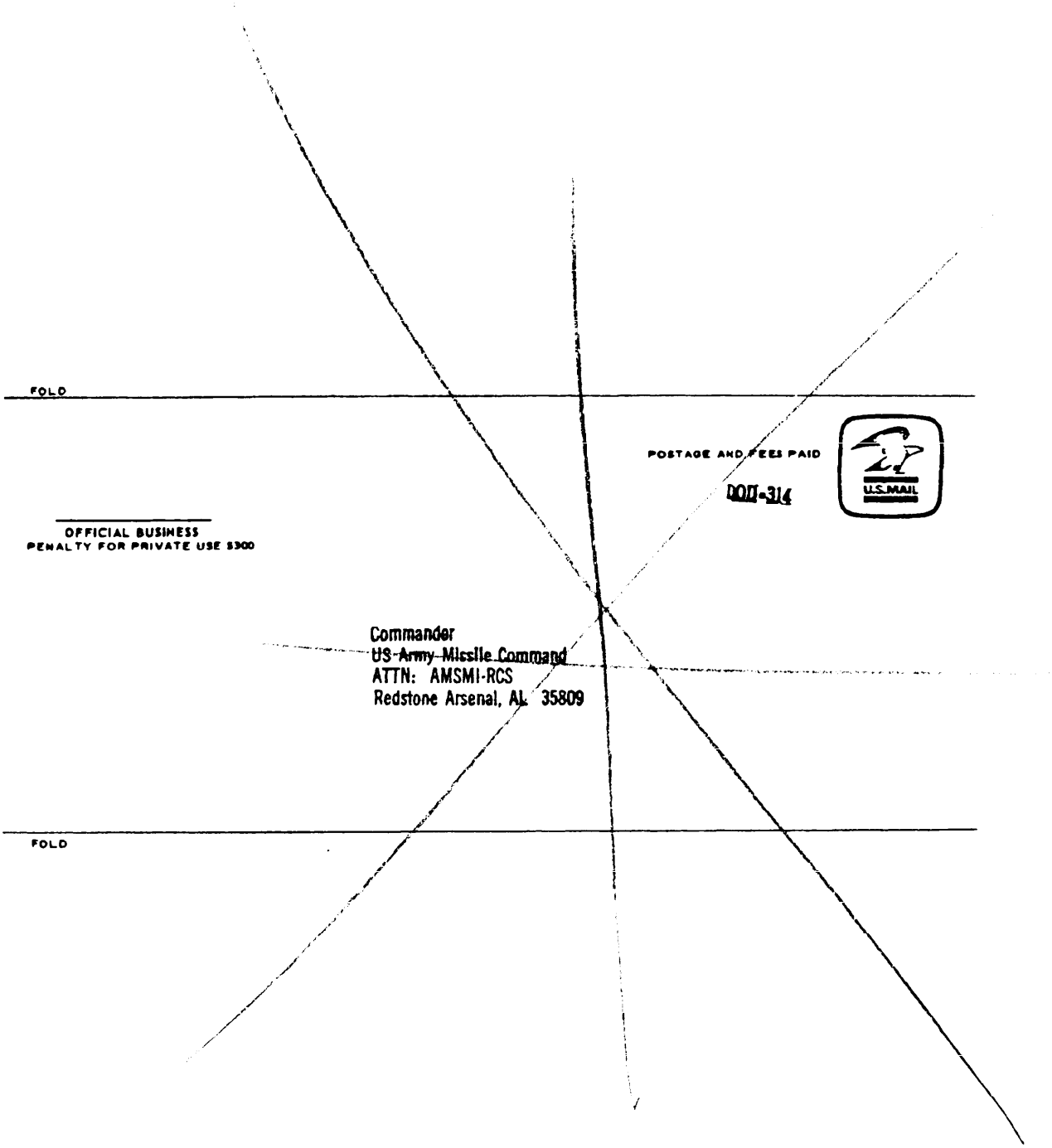
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